



Horizontal combinations of online and offline approximate dynamic programming for stochastic dynamic vehicle routing

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Abstract

Stochastic and dynamic vehicle routing problems gain increasing attention in the research community. In these problems, routing plans are dynamically updated based on realizations of stochastic information. Due to the complexity of the corresponding Markov decision processes (MDPs), the calculation of optimal policies for these problems is usually not possible and researchers draw on heuristical methods of approximate dynamic programming (ADP). These methods use simulation to approximate the value of a state and decision in the MDP. The simulations are either conducted offline or online. Offline methods such as value function approximations (VFAs) generally neglect the full detail of the state space due to aggregation. Online methods such as rollout algorithms (RAs) are often not able to capture decision and transition space sufficiently due to runtime limitations. In this paper, we alleviate this tradeoff by combining two methods of ADP, an online RA and an offline VFA in two ways. In addition to the integration of the VFA as a base policy into the online RA to strengthen the RA's simulations, we also limit the RA's simulation horizon, estimating the remaining reward-to-go again via the VFA. For two stochastic dynamic routing problems from the literature, we show how this combination outperforms state-of-the-art solutions while simultaneously reducing the required time for online calculations.

Keywords Approximate dynamic programming · Rollout algorithm · Value function approximation · Dynamic vehicle routing · Stochastic requests · Multi-period

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1 Introduction

Recent developments in business models and communication technology lead to a changing nature in vehicle routing problems. In many applications, dispatchers subsequently update their plans in a decision process based on new information (Gendreau et al. 2016). These applications are particularly common in volatile urban environments and comprise traditional applications like courier services, but also emerging business models from the fields of shared mobility or same-day delivery (Ulmer 2017). As Savelsbergh and Van Woensel (2016) state, providing suitable customer service in these applications “is becoming exceedingly difficult due to the increased complexity, dynamics, and uncertainty” of the business models. Suitable methods of operations research need to acknowledge “the importance of appropriately modeling dynamic events and simultaneously incorporating information about the uncertainty of future events” (Speranza 2018). This is conducted in the scientific field of stochastic dynamic vehicle routing (SDVR).

1.1 Markov decision process

To address the stochastic and dynamic nature in SDVR, the applications are modeled as Markov decision process (MDP, Ulmer et al. 2017). An MDP represents a sequence of decision states connected by decisions and realizations of the stochastic information process. A solution of an MDP is a policy. A policy assigns a decision to each particular state. An optimal policy maximizes the expected rewards (or minimizes expected costs respectively). Each decision of an optimal policy satisfies the Bellman Equation maximizing the sum of immediate and expected future rewards. The expected future rewards are summarized in the *value* of a state and decision combination. The values and therefore the optimal policy of small MDPs can be determined by means of recursive calculation. This is not possible for the complex MDPs representing real-world applications in SDVR due to the Curses of Dimensionality (Powell 2011). In SDVR, state, decision, and transition spaces are vast. A state often comprises information such as the current point of time, the vehicles’ and customers’ positions, or inventory-levels of vehicles and customers. Decisions are made about the routing, more specific, the assignment and sequencing of customers. The transitions may comprise new customer requests (Voccia et al. 2017), the realization of stochastic service times (Zhang et al. 2018), demands (Goodson et al. 2016), or travel times (Schilde et al. 2014). In essence, a recursive calculation of the values in SDVR is generally not possible.

1.2 Approximate dynamic programming

Instead of exactly calculating the values, methods of approximate dynamic programming (ADP) draw on simulation to *approximate* the values of a particular state. The majority of ADP-methods either simulates *offline* in a learning phase or simulates *online* being in a particular decision state. The most common offline ADP-method is value function approximation (VFA). A VFA approximates the value of a state based on a small set of state features such as the point of time. Due to basically unlimited

calculation time available, the VFA is able to consider the dimensions of decision and transition spaces in high detail. However, the state space is aggregated to the small set of features and many state details such as vehicle locations or planned routes are neglected (Goodson 2010; Ulmer et al. 2018b).

To capture these details, an online simulation is necessary. These methods are called rollout algorithms (RAs). For a general overview on RAs, we refer the interested reader to the recent survey by Goodson et al. (2017). An RA starts in a state and simulates potential future realizations of the MDP to estimate the value for this particular state. In the simulation, transitions are sampled and decisions are derived by a base policy. In SDVR, the online calculation time accessible in a state is highly limited. Only a few simulations are possible and the base policy is usually a simple rule-of-thumb. Thus, RAs are able to incorporate all state details but the consideration of decision and transition space is limited because of the small number of simulation runs. This leads to insufficient approximation, particularly, when given long simulation horizons in SDVR. This phenomenon is often observed in the literature. As examples, Voccia et al. (2017) experience that “1 and 2 h horizons perform worse than the shorter 1/2-h sampling horizon.” Their experiments “indicate that longer sampling horizons require a larger number of scenario samples.” In essence, a long horizon may result in inferior state evaluation and decision making. Experiments from Ghiani et al. (2009) for a dynamic routing problem suggest that increasing the simulation horizon “would not result in any further objective function improvement, while causing an increase in the average CPU time per request.” As a result, many approaches limit the horizon of the online simulation and only simulate short-term. Thus, these methods neglect the long-term impact of decision making.

1.3 Combining online and offline ADP

In this paper, we present a novel way to combine methods of approximate dynamic programming allowing detailed online simulations in tandem with the consideration of the decision’s long-term impact. To this end, we reinforce online simulations with VFAs in two ways. First, as Ulmer et al. (2018a) and Li and Womer (2015) propose, our base policy draws on the VFA values. Second, to account for the long horizons, we introduce the idea of splitting the simulations of the MDP horizontally. The first part of the horizon is approximated by online simulations. The remaining horizon is approximated by the VFA values. We denote our policy *VFA-based limited horizon rollout algorithm* (V-LHRA).

To the best of our knowledge, this procedure has not been applied in the SDVR-literature. However, the idea of horizontally combining online simulations with VFAs is mentioned in the literature (Powell and Meisel 2016). The general idea of evaluating the outcome of an online simulation with VFAs in SDVR is briefly sketched in Powell et al. (2012). Our method is also similar to variants of Monte Carlo tree search, often applied to generate policies for complex games with long horizons such as *Go* (Browne et al. 2012).

We show the benefits of this method for two dynamic vehicle routing problems with stochastic service requests from the literature differing in the horizon’s length.

Stochastic service requests are often observed in applications from the fields of courier services, technician services, or installation services where service is requested while the service vehicle is already on the road. For these applications, the vehicle is dynamically routed in response to new customer requests during the day (Thomas 2007). The first problem is limited to one period or day. Decisions are made about the acceptance or rejection of new customer requests as well as the dynamic routing of the service vehicle. The second problem, the *multi-period dynamic vehicle routing problem with stochastic customer requests* (MDRPSR) is a generalization of the first problem (Ulmer et al. 2018c). It reflects applications where customers cannot be rejected but are served either the same day or the next day. Thus, the problem comprises several periods. Within each period, customers request service. Decisions are made about the acceptance of customers for same-day service or postponement of customers and the routing of the vehicle in the current period. The postponed customers form the initial customers of the next period. To analyze the impact of the horizontal combination of RA and VFA, we compare our method to the combination proposed in Ulmer et al. (2018a) as well as VFA and online simulations applied in isolation. We show that our method increases solution quality and, at the same time, reduces of the required online calculation times. Further, we show that our method is particularly beneficial for problems with long horizons.

1.4 Contributions and outline

This paper makes the following important contributions. Methodologically, we present and analyze a strong tunable tool for stochastic dynamic routing problems with a high number of decision points and highly limited online calculation time. Our method can be applied to many problems if an offline VFA is accessible. To the best of our knowledge, a combination of limited horizon online ADP and offline VFA has never been applied in SDVR. We show the performance for two dynamic routing problems with stochastic requests from the literature. Our methods provide high quality solutions for a wide range of instance settings outperforming current state-of-the-art benchmark policies in both solution quality and runtime efficiency. We further show how a suitable simulation horizon is strongly connected to the online runtime accessible.

The paper is outlined as follows. In Sect. 2, we present the literature relevant to the presented methodology and the problem area. We model the general MDRPSR as a Markov decision process in Sect. 3 and define our method and the benchmark approaches in Sect. 4. We present our computational studies in Sect. 5. The paper concludes with a summary and an outlook in Sect. 6.

2 Literature

In the literature review, we focus on work applying anticipatory optimization techniques to dynamic routing problems with stochastic requests and on work regarding dynamic multi-period vehicle routing. For a general overview on dynamic routing problems, we refer to Ulmer (2017). For an overview on deterministic multi-period

vehicle routing problems, the interested reader is referred to Archetti et al. (2015). Multiple trips to the depot or decision making over multiple periods are further considered in other disciplines like dynamic routing with stochastic demands but without stochastic requests (Goodson et al. 2016), or scheduling without considering routing (Chen et al. 2017).

The literature is presented in Table 1. We classify the literature with respect to the model and the solution method. We analyze whether the model reflects multiple periods (“✓” in the multi-period column) or vehicles performing multiple routes within a single period, indicated by “(✓)” in the multi-period column. We further analyze if new requests can be integrated in the current routes or only in subsequent routes. We classify the approaches with respect to their anticipation. An approach is called “anticipatory” if it integrates stochastic information in the decision making process. In case an approach is anticipatory, we determine when significant calculations are conducted.

Beside Ulmer et al. (2018c), the work on dynamic multi-period routing is limited. Angelelli et al. (2009) present a problem similar to the MDRPSR. However, they allow rejections. Acceptances, postponements, and rejections are not permanent, but can be

Table 1 Literature classification

Literature	Model		Approach		
	Multi-period	Integration	Anticipation	Online	Offline
Bent and Van Hentenryck (2004)		✓	✓	✓	
Ichoua et al. (2006)		✓	✓	✓	
Thomas (2007)		✓		✓	✓
Ghiani et al. (2009)		✓	✓	✓	
Angelelli et al. (2009)	✓	✓			
Wen et al. (2010)	✓			✓	
Meisel (2011)		✓		✓	✓
Azi et al. (2012)	(✓)			✓	
Ferrucci et al. (2013)		✓	✓	✓	
Albareda-Sambola et al. (2014)	✓		✓	✓	
Klapp et al. (2018b)	(✓)		✓	✓	
Klapp et al. (2018a)	(✓)		✓	✓	
Sarasola et al. (2016)		✓	✓	✓	
Tirado and Hvattum (2017)	(✓)		✓	✓	
Ulmer et al. (2018c)	✓	✓		✓	✓
Ulmer et al. (2018d)	(✓)		✓		✓
Vonolfen and Affenzeller (2016)	(✓)		✓	✓	
Ulmer et al. (2018b)		✓	✓		✓
Ulmer et al. (2018a)		✓	✓	✓	✓
Voccia et al. (2017)	(✓)		✓	✓	
MDRPSR, V-LHRA	✓	✓	✓	✓	✓

changed over time. Angelelli et al. (2009) apply short-term priority rules based on the tour duration and the number of requests postponed without considering stochastic information about potential future requests. Wen et al. (2010) and Albareda-Sambola et al. (2014) present problems where routes are determined once every period and an integration of new requests in the current period's tour is not feasible. Customers can only be served in the periods after the request.

Multiple subsequent routes are also addressed in research on same-day delivery. Voccia et al. (2017), Klapp et al. (2018b), Ulmer et al. (2018d) present work where vehicles deliver packages from a depot to dynamically requesting customers. Before serving a customer, the vehicle returns to the depot to pick up the order. Hence, the vehicles perform a sequence of routes over the course of the day. These problems differ from the MDRPSR in two ways. First, the duration for each route is flexible and not fixed by the period's working hours. Second, as in Wen et al. (2010) and Albareda-Sambola et al. (2014), requests are not integrated in the current but only in the subsequent route. Similar work is presented by Tirado and Hvattum (2017) and Vonolfen and Affenzeller (2016). They consider stochastic and dynamic pickup and delivery problems and solve the problem by combining sampling with local search-heuristics and stochastic waiting strategies, respectively.

In the following, we analyze the work with respect to anticipation. Since the MDRPSR with only one period is a dynamic single-period vehicle routing problem, we also analyze work on dynamic single-period vehicle routing with stochastic requests. The anticipatory approaches can be classified by the point of time in which the majority of calculation is conducted, either online in the particular decision state, or offline before the algorithm is applied.

Offline algorithms are highly valuable since they determine decisions immediately in an online decision state. However, they often limit the set of possible decisions (Thomas 2007; Meisel 2011) and/or only consider states on an aggregated level (Ulmer et al. 2018b). For the MDRPSR, Ulmer et al. (2018c) present an offline VFA approximating values with respect to temporal attributes like point of time and free time budget per period left. Offline VFAs are usually not able to incorporate every state's details like customers' or vehicle's position. This information is only accessible in the online execution. To analyze the impact of the V-LHRA's online simulations, we select the VFA as a benchmark policy.

In contrast to offline approaches, online algorithms face limited calculation time, but are able to consider every aspect of the decision state. However, online algorithms often capture only a limited horizon neglecting later developments (Ghiani et al. 2009) or may not consider multiple future adaptations of plans explicitly (Bent and Van Hentenryck 2004; Ichoua et al. 2006; Ferrucci et al. 2013; Albareda-Sambola et al. 2014; Voccia et al. 2017; Sarasola et al. 2016; Tirado and Hvattum 2017; Vonolfen and Affenzeller 2016). Some online approaches consider the entire MDP-horizon (Ulmer et al. 2015; Klapp et al. 2018b). In every decision point, Klapp et al. (2018b) solves the relaxed sub-MDP via linear programming. Ulmer et al. (2015) apply a rollout algorithm. Within the simulation, the RA draws on a base policy to determine plan updates within the simulations to evaluate current decisions. Since the base policy usually is applied several times per simulation run, Ulmer et al. (2015) apply a simple myopic decision rule within the simulations. As Powell and Meisel (2016) summarize,

online approaches often provide detailed, but local anticipation of the MDP, offline approaches provide global, but aggregated anticipation. To combine advantages of offline and online anticipation, Ulmer et al. (2018a) integrate an offline VFA into the online rollout algorithm for the single-period version of the MDRPSR to a VFA-based rollout algorithm (V-RA). They show how this combination is superior to the approaches applied in isolation.

The V-RA captures the entire horizon of the MDP with online simulations. However, the MDRPSR contains a multiple number of decision points over the periods. A simulation until the end of the horizon may require significant calculation time and may provide distorted results because the simulated requests may strongly differ from the actual request realization. The impact of this difference increases over the horizon. To alleviate this issue, we generalize the V-RA to the V-LHRA by limiting the online simulation horizon and estimate the remainder of the horizon by the more reliable offline VFA. We show the benefit by comparing our V-LHRA with a version of the V-RA adapted to multi-period settings.

3 The MDRPSR

In this section, we formulate the MDRPSR and model it as Markov decision process.

3.1 Problem formulation

In the following, we give a problem formulation for the MDRPSR and introduce the mathematical notation. The MDRPSR reflects a variety of service applications such as courier, installation, or technician services where customers request service over the course of the day and expect fast service either today or the next day. On the operational level of these problems, service resources are generally given. The provider's goal is to utilize these resources to service as many customers as possible the same day. This leads to an increase in customer satisfaction and to a reduction of workload for the following days. Mathematically, we formulate the MDRPSR as follows:

Given a finite number of periods $\Delta = \{1, \dots, \delta_{\max}\}$, a vehicle serves customers $\mathcal{C}$ in the service area $\mathcal{A}$. Each period contains a sequence of points of time $T = (0, 0 + \bar{t}, \dots, t_{\max})$ with basic time unit $\bar{t}$. In every period, the vehicle starts the tour in $t = 0$ at depot D with location $l^D \in \mathcal{A}$ and returns within a time limit $t_{\max}$. In the first period, a set of early request customers (ERCs) $\mathcal{C}^e$ is known and must be served. During and over the periods, new requests from the set of late request customers (LRCs) $\mathcal{C}^l$ arrive stochastically over time. Their locations in the area follow a known spatial probability distribution. Notably, these requests are unknown before their time of request. The overall set of customers is $\mathcal{C} = \mathcal{C}^e \cup \mathcal{C}^l$. Each customer $C \in \mathcal{C}$ is defined by a vector $C = (l^c, \delta^c, t^c)$ containing the location $l^c \in \mathcal{A}$, the request period $\delta^c \in \Delta$, and the time the request is placed $t^c \in T$. The travel time between two locations $l_1, l_2 \in \mathcal{A}$ is defined by function $d(l_1, l_2)$.

When arriving at a customer, the dispatcher selects the subset of new requests to accept for the current period, thus postponing the remaining requests to the next period. The vehicle then travels to the next customer. These decisions are permanent. Postponed requests must be served in the following period. Postponements in $\delta_{\max}$ are without consequences. For this operational problem, the resources are fixed and the drivers are paid. Thus, the objective is to maximize the service level, i.e., the number of requests accepted for same-period service, over all periods. Because of the stochastic nature of the problem, in a few cases, the initial tour duration visiting the ERCs or the postponed customers of the previous period may violate the time limit. In this case, the initial customers are visited and all following LRCs are postponed leading to no acceptances in this period.

Notably, the MDRPSR is a generalization of the single-period dynamic vehicle routing problem considered, amongst others, by Thomas (2007) and Ulmer et al. (2018a). In the single-period version, customers are not postponed but rejected. The MDRPSR can be transferred to this problem by setting $\delta_{\max} = 1$. With increasing number of periods $\delta_{\max}$, the MDRPSR further converges to an infinite horizon problem. In that case, discounting of rewards may be required.

3.2 Markov decision process

We model the MDRPSR as a route-based Markov decision process (MDP, Ulmer et al. 2017). In an MDP, a set of decision points $k = 1, \dots, K$ is given. Decision point k is connected to the subsequent decision point by the sequence of state S_k , a decision x , a post-decision state S_k^x and a stochastic transition ω_k leading to the next decision point. In a route-based MDP, the decision space considers routing plans. We select a route-based MDP because it directly connects our solution method to the model. For an MDP-example of the MDRPSR, the interested reader is referred to Ulmer et al. (2018c).

Decision states

A decision point k occurs in two cases: at the beginning of each period and when the vehicle finished serving a customer. A state S_k in state space $\mathcal{S}$ for this problem is defined by the current period δ_k , the point of time t_k , the vehicle's location $l_k^v \in \mathcal{A}$, the set of customers C_k^δ to serve in this period, and the set of customers $C_k^{\delta+1}$ to serve in the following period, and the routing plans in the current and next period θ_k^δ and $\theta_k^{\delta+1}$ respectively. A routing plan is a sequence through the customers in the period starting at the current location and ending at the depot $\theta_k^\delta = (l_k^v, C_1^\delta, \dots, C_m^\delta, D)$ and $\theta_k^{\delta+1} = (D, C_1^{\delta+1}, \dots, C_n^{\delta+1}, D)$. In $\delta_{\max}$, the set of postponed customers and the next-period's tour are empty, $C_k^{\delta+1} = \emptyset$ and $\theta_k^{\delta+1} = \emptyset$. Further, S_k contains a set of new requests $C_k^r \subset C^l$. In essence, a state is defined by $S_k = (\delta_k, t_k, l_k^v, C_k^\delta, C_k^{\delta+1}, \theta_k^\delta, \theta_k^{\delta+1}, C_k^r)$. The initial state S_0 is defined by $S_0 = (1, 0, l^D, C^e, \emptyset, (D), (D), \emptyset)$. The termination state S_K is defined by $S_K = (\delta_{\max}, t_{\max}, l^D, \emptyset, \emptyset, (D), \emptyset, \emptyset)$.

Decisions

In every decision point k , decisions x are made about the subset $\mathcal{C}_k^a$ of requests $\mathcal{C}_k^r$ to accept for same-day service and an update of tours θ_k^δ and $\theta_k^{\delta+1}$. Given n new requests, up to 2^n potential acceptance decisions are possible. The set of potential routing updates depends on the existing, the newly accepted, and the postponed customers. The routing update of the current period induces the next customer location to visit $l_k^{\text{next}} \in \{l_k^v\} \cup \mathcal{L}_k^x \cup \{l^D\}$ with $\mathcal{L}_k^x \in \mathcal{A}$ the locations of $\mathcal{C}_k^{\delta,x} = \mathcal{C}_k^\delta \cup \mathcal{C}_k^a$. Let customer $\mathcal{C}_k^{\text{next}}$ be associated with the location l_k^{next} . In case $l_k^{\text{next}} = l_k^v$, the vehicle idles for one time unit, i.e., $d(l_k^v, l_k^v) := \bar{t}$. The reward $R(S_k, x) = |\mathcal{C}_k^a|$ of decision x is the number of accepted customers. A decision is feasible if θ_k^δ contains all $\mathcal{C}_k^{\delta,x}$ and allows to return to the depot D before $t_{\max}$. In the case that the tour duration of θ_k^δ exceeds the time limit, a decision is only feasible if no customer is accepted, $\mathcal{C}_k^a = \emptyset$. The overall duration $\bar{d}(\theta_k)$ is defined by the sum of single travel times $\bar{d}(\theta_k) = d(l_k^v, l^{c1}) + d(l^{c1}, l^{c2}) + \dots + d(l^{ci}, l^D)$ with $i = |\mathcal{C}_k^{\delta,x}|$.

We split the transition in the deterministic transition by decision x and the stochastic transition by a realization ω of the information space. The combination of state S_k and decision x results in a (known) post-decision state (PDS) S_k^x from the post-decision state space $\mathcal{S}^x$. The dimensions of the PDS are similar to the state except for the set of requests. The PDS is defined by $S_k^x = (\delta_k, t_k, l_k^v, \mathcal{C}_k^{\delta,x} \cup \mathcal{C}_k^a, \mathcal{C}_k^{\delta+1,x}, \theta_k^{\delta,x}, \theta_k^{\delta+1,x})$ with $\mathcal{C}_k^{\delta+1,x} = \mathcal{C}_k^{\delta+1} \cup \mathcal{C}_k^r \setminus \mathcal{C}_k^a$.

Transitions

There are two types of transitions from a post-decision state to a new state. The first type is the transition within the period, the other transition is inter-periodically. The transition within the period reflects the vehicle traveling to the next customer and potentially a set of new requests. The transition to the next period reflects the end of the current day and the beginning of the next day. Mathematically, the transitions are modeled as follows:

Within the period, the realization of the information space ω_k from PDS S_k^x results in a state S_{k+1} with $\delta_{k+1} = \delta_k$, $t_{k+1} = t_k + d(l_k^v, l_k^{\text{next}})$, $l_k^v = l_k^{\text{next}}$, $\mathcal{C}_{k+1}^\delta = \mathcal{C}_k^{\delta,x} \setminus \{\mathcal{C}_k^{\text{next}}\}$, and $\mathcal{C}_{k+1}^{\delta+1} = \mathcal{C}_k^{\delta+1,x}$. Further, the routing plans are updated as follows: $\theta_{k+1}^\delta = \theta_k^{\delta,x} \setminus \{\mathcal{C}_k^{\text{next}}\}$, and $\theta_{k+1}^{\delta+1} = \theta_k^{\delta+1,x}$. The inter-period transition occurs if $S_k^x = (\delta_k, t_{\max}, D, \emptyset, \mathcal{C}_k^{\delta+1,x}, \theta_k^{\delta,x}, \theta_k^{\delta+1,x})$ with $\delta_k < \delta_{\max}$. The new state S_{k+1} is then defined by $S_{k+1} = (\delta_k + 1, 0, l^D, \mathcal{C}_k^{\delta+1,x}, \emptyset, \theta_K^{\delta+1,x}, (D), \mathcal{C}_{k+1}^r)$. The tour of $\delta + 1$, $\theta_K^{\delta+1,x}$ becomes the new current tour, and the new tour $\theta_0^{\delta+2} = (D)$ only contains the depot.

Objective

A solution for a Markov decision process is a policy π from the overall set of policies Π . A policy π is a sequence of decision rules $(X_0^\pi, X_1^\pi, \dots, X_K^\pi)$ for every decision point $k = 1, \dots, K$. Each decision rule $X_k^\pi(S_k)$ specifies the decision $x \in X(S_k)$ to

select when the process occupies state S_k . The objective is to find an optimal decision policy $\pi^* \in \Pi$ leading to the highest expected sum of rewards (Puterman 2014). An optimal policy maximizes the sum of expected rewards as stated in Eq. (1):

$$\pi^* = \arg \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{k=0}^K R(S_k, X_k^\pi(S_k)) | S_0 \right]. \quad (1)$$

For the problem at hand, we omit potential discounting of the reward. However, the proposed method works as well with discounted rewards. We denote the MDP starting in S_k as “sub-MDP”. The optimal decision x_k^* in a specific state S_k can be derived by solving the sub-MDP as shown in Eq. (2):

$$x_k^* = \arg \max_{x \in X(S_k)} \left\{ R(S_k, x) + \mathbb{E} \left[\sum_{j=k+1}^K R(S_j, X_j^\pi(S_j)) | S_k^x \right] \right\}. \quad (2)$$

In decision step k , the decision is selected that maximizes the sum of immediate and expected future rewards based on *Bellman's Equation*. The expected future rewards are also called the value $V(S_k^x)$ of post-decision state S_k^x . For the MDRPSR, the value of a post-decision state is the expected number of acceptances over all following decision points, i.e., the acceptances in the remaining time of the current period and the expected acceptances over the following periods.

4 VFA-based limited horizon rollout algorithm

Determining an optimal solution of the sub-MDP starting in S_k is not possible due to the “Curses of Dimensionality” in several dimensions. For the MDRPSR, the number of states and transitions is vast, the decision space large, and, in particular, the problem horizon long. In this section, we present a VFA-based limited horizon rollout algorithm (V-LHRA) and describe how the V-LHRA addresses the individual curses. To analyze the performance, we further present a set of benchmark policies.

4.1 Reducing the curses

The V-LHRA applies reduction techniques in all dimensions. To reduce the number of transitions (also called “curse of expectation”), we apply a post-decision rollout algorithm sampling trajectories from state S_k . For a general overview on RAs, the interested reader is referred to Goodson et al. (2017). Since many decisions have to be made within the simulation runs while calculation time is highly limited, we draw on an offline VFA as base policy as proposed in Ulmer et al. (2018a).

For the MDRPSR, the number of decision points is large since they occur at every visited customer over all periods. Thus, we reduce the RA-horizon by limiting the simulation horizon to h decision points, therefore only simulating until post-decision state S_{k+h}^x . To estimate the reward-to-go after this state, again, we draw on the offline

VFA. Finally, the combination of routing and subset selection leads to a large number of potential decisions. Since Ulmer et al. (2018b) shows that acceptance decisions have the greatest impact on the solution quality, we focus on subset selection and apply a cheapest insertion routing heuristic (Rosenkrantz et al. 1974). For a detailed description of the routing, we refer the reader to Sect. A.1 in the “Appendix”.

4.2 Procedure: V-LHRA

In the following, we introduce the V-LHRA. Each potential acceptance decision in combination with the routing heuristic leads to a potential post-decision state. Policy V-LHRA evaluates each of these post-decision states with respect to the expected number of future same-period services and selects the decision via the (approximate) Bellman Equation. In the following, we describe the procedure of how the estimation is obtained. The detailed algorithm is described in Sect. A.2 of the “Appendix”. Notably, the V-LHRA requires the existence of a value function approximation. A VFA is a function $\tilde{V} : \mathcal{S}^x \rightarrow \mathbb{R}$ and maps post-decision states to real numbers. Thus, a VFA provides a value $\tilde{V}(S_k^x)$ for every post-decision state.

Figure 1 describes the process of the V-LHRA given a particular post-decision state S_k^x . The post-decision state is represented by the circle on the left. From the post-decision state, m sampled *online* transitions $\omega_i^k, i = 1, \dots, m$ lead to a set of m (simulated) new decision states $S_{k+1}(\omega_i^k)$ in $k + 1$. The decision in every state is conducted by the VFA-policy represented by $X^b(S_{k+1}(\omega_i^k))$ via the approximate Bellman Equation:

$$X^b(S_{k+1}(\omega_i^k)) = \arg \max_{x \in X(S_{k+1})} \left\{ R(S_{k+1}(\omega_i^k), x) + \tilde{V}(S_{k+1}^x(\omega_i^k)) \right\}. \quad (3)$$

Notably, the decisions of the base policy are applied only within the simulation and not to the actual implementation. The application leads to a (simulated) reward $R_{k+1}(S_{k+1}(\omega_k), X^b(S_{k+1}(\omega_i^k)))$, in Fig. 1 simplified by R_{k+1} , and the next transition ω_i^{k+1} is sampled. This continues until post-decision state S_{k+h}^x is reached. The value for each realization is then the sum of accumulated rewards **plus** the VFA-value $\tilde{V}(S_{k+h}^x)$ of the last simulated post-decision state. This is indicated by the dashed line of the right side of Fig. 1. The overall value $\hat{V}(S_k^x)$ for each post-decision state is

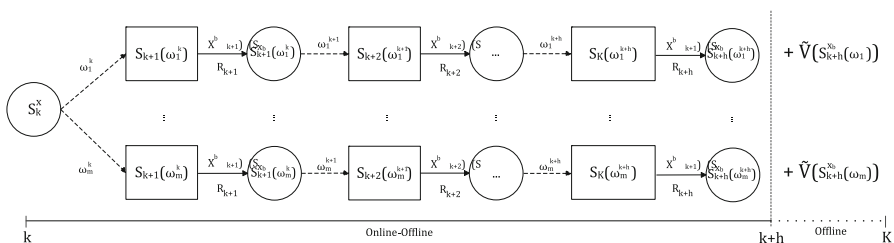


Fig. 1 The VFA-based limited horizon rollout algorithm

therefore

$$\widehat{V}(S_k^x) = \frac{1}{m} \sum_{i=1}^m \left(\sum_{j=k+1}^{k+h} R_j \left(S_j(\omega_i^{j-1}), X^b \left(S_j(\omega_i^{j-1}) \right) \right) + \tilde{V} \left(S_{k+h}^x(\omega_i^{k+h-1}) \right) \right). \quad (4)$$

The approximate value of a post-decision state is the average over the realized rewards of the m simulations. The reward for each simulation is the sum of the online rewards over the h decision points simulated online plus the offline rewards approximated by the offline VFA. The approximate values for each post-decision state in decision point k can then be used to derive the actual decision x_k^* , again, via the approximate Bellman Equation:

$$x_k^* = \arg \max_{x \in X(S_k)} \{ R(S_k, x) + \widehat{V}(S_k^x) \} \quad (5)$$

Decision x_k^* is then applied and the MDP continuous to the next decision point $k + 1$.

4.3 Tuning and offline VFA

In this section, we describe the tuning of V-LHRA for the MDRPSR. We test the V-LHRA for the single- and multi-period version of the MDRPSR. We vary the online horizon of the V-LHRA for the single-period version between $h = 5, 10, \dots, 80$ and for the multi-period version between $h = 10, 20, \dots, 220$ since preliminary test revealed a usual number of less than 70 and 200 decision points, respectively. We run 200 tuning runs for each instance setting and select the horizon providing the highest solution quality (compare Tables 5, 6 in the “Appendix”). We apply the V-LHRA with $m = 2, 16, 128$ online simulations without parallelization. All methods are coded in Java and executed on 2.4 GHz AMD Opteron dual-core processors with 8GB of RAM.

In the following, we describe how we obtain a VFA for the MDRPSR. Since we apply V-LHRA to both the single- and multi-period version of the MDRPSR, we draw on two different, non-parametric VFAs. A non-parametric VFA approximates values individually without any assumptions of the value **function**. The two VFAs estimate the values for highly aggregated states to allow a storage of the values. The aggregation scheme follows Ulmer et al. (2018b) and aggregates states with respect to temporal attributes such as the point of time and the free time budget left to serve additional customers. The free time budget is the amount of time the vehicle arrives early at the depot if it follows the current routing plan. The aggregation schemes for single- and multi-period version of the MDRPSR differ. In the single-period version, the VFA aggregates states to two-dimensional vectors of point of time and free time budget. In the multi-period version, the VFA is applied aggregating states to four-dimensional vectors containing the period, the point of time, the free time budget in the current, and the free time budget in the next period.

The values for each vector are then approximated by means of approximate value iteration (AVI). For a detailed definition of AVI, the interested reader is referred to (Powell 2011, p.127ff). The procedure starts with initial values for each vector. These values induce a policy with respect to the approximate Bellman Equation. Then, AVI

frequently simulates problem realizations by applying the current policy. After each simulation run, the values are updated according to the realized values and a new policy is induced. This procedure continues until the maximal number of simulation runs is reached. We tune both offline VFAs with 1 million approximation runs and update the values with the running average. To improve the approximation process, we additionally apply a dynamic disaggregation scheme, the dynamic lookup table, with disaggregation thresholds 1.5 for the single-period version and 3.0 for the multi-period version (compare Ulmer et al. 2018b,c). The final VFAs are then utilized in the according V-LHRA.

4.4 Benchmark policies

We compare the V-LHRA to five benchmark heuristics. First, to analyze the anticipation quality, we compare to a myopic policy. This policy sets the value $\widehat{V}(S_k^x)$ to zero for every post-decision state. This results in the acceptance of the maximal feasible subset in each decision point.

We then define benchmark policies to analyze V-LHRA with respect to two dimensions, the VFA base policy and the limited horizon. The three benchmark policies are depicted in Table 2. To analyze the impact of the VFA base policy, we apply RA with a myopic base policy on the unlimited and limited horizon: M-RA and M-LHRA. The value at the end of the limited horizon of M-LHRA is approximated by the myopic base policy and is, therefore, zero. To analyze the impact of the limited horizon, we apply RAs with unlimited horizon for both base policies: M-RA and V-RA. To analyze how the online component improves the offline approximation, we additionally apply the offline VFA in isolation as depicted in Eq. (3). The tuning of the benchmark policies follows the tuning of V-LHRA.

The six policies can be summarized as follows:

1. V-LHRA: This is the proposed policy. This policy simulates a limited horizon using the offline VFA as base policy. The policy then estimates the remaining rewards after the horizon with the VFA-value.
2. M-LHRA: This benchmark policy simulates a limited horizon with a myopic base policy using the observed rewards to evaluate the post-decision states.
3. V-RA: This benchmark policy simulates the entire horizon with the offline VFA. This is the current state-of-the art policy for the single-period MDPRSR.
4. M-RA: This benchmark policy simulates the entire horizon with a myopic base policy.
5. VFA: This benchmark policy evaluates post-decision states with the offline VFA-values. This is the current state-of-the art policy for the multi-period MDPRSR.
6. myopic: This benchmark policy accepts the largest feasible subset in every decision point. In case of several potential subsets, the policy minimizes the increase in tour duration of the current period's routing.

Table 2 Dimensions of the benchmark policies

	Offline-online	Online
Limited	V-LHRA	M-LHRA
Unlimited	V-RA	M-RA

5 Computational evaluation

In this section, we define the instances and compare the solution quality of the policies, first for the single-period version of the MDRPSR, then for the multi-period version. We finally analyze the impact of the limited horizon on the solution quality and runtime.

5.1 Instances

The instances are derived from Ulmer et al. (2018a) and extended to multi-period settings. The time limit per period is set to $t_{\max} = 360$ min, the basic time unit is $\bar{t} = 1$ minute. The depot is located in the center of the service area. The vehicle travels with a speed of 20 km per hour. We vary the service area size between *small* (10 km $\times$ 10 km) and *large* (15 km $\times$ 15 km). We further examine the impact of different spatial customer distributions. We generate uniformly distributed locations (U), customers accumulated in three (3C) and two clusters (2C). For the details of the distributions, we refer to the “Appendix”. We finally vary the number of dynamic requests setting the degree of dynamism to 0.5 and 0.75 and the expected number of requests to 100 in the first period reflecting quantities of parcel transportation. Notably, the vehicle does not serve all request in one period, but only a percentage while postponing the rest to the next period. A degree of dynamism of 0.5 means that 50 ERCs and 50 LRCs are expected in the initial period. Then, in the next periods the number of ERCs depend on the previous postponements and 50 LRCs are expected per period. In the case of 0.75, 25 ERCs and 75 LRCs are expected in the first period and 75 LRCs in the following periods. Again, only a percentage of these 75 requests is accepted while the rest is postponed to the next period.

We create instances for the special single-period case $\delta_{\max} = 1$ to compare the results with the single-period problem. We further set $\delta_{\max} = 3$ for the multi-period case. Due to runtime limitations of our benchmark policies, we omit tests for more than three periods. However, our results indicate that limiting the online simulation horizon becomes even more effective when the number of periods is increased. Overall, we consider 24 different instance settings. We generate 200 realizations per instance setting.

5.2 Solution quality

In this section, we present and analyze the solution qualities for the defined instances. We define the solution quality $Q(\pi, i)$ of policy π as the average percentage of same-day services per instance setting i . The results are depicted in Table 3. To measure the significance of the results, we further present the standard error for each result.

Table 3 Results: percentage of same-day services and standard errors

Distribution	2C		2C		2C		3C		3C		3C		U		U		U	
	0.5	Small	0.75	Small	0.5	Large	0.75	Large	0.5	Small	0.75	Large	0.5	Small	0.75	Small	0.5	Large
Area size																		
$\delta_{\max} = 1$																		
M	50.6 ± 0.6	53.9 ± 0.4	16.2 ± 0.6	36.7 ± 0.4	75.4 ± 0.7	72.0 ± 0.5	55.2 ± 0.7	56.1 ± 0.4	83.1 ± 0.7	76.6 ± 0.5	62.8 ± 0.6	63.5 ± 0.5						
VFA	57.8 ± 0.6	59.9 ± 0.4	22.5 ± 0.7	46.3 ± 0.4	74.4 ± 0.7	71.6 ± 0.5	59.4 ± 0.7	60.0 ± 0.4	82.4 ± 0.7	76.1 ± 0.5	64.9 ± 0.6	66.8 ± 0.5						
M-LHRA (2)	54.1 ± 0.6	55.5 ± 0.4	19.9 ± 0.6	40.4 ± 0.4	75.1 ± 0.7	70.8 ± 0.5	57.5 ± 0.6	57.5 ± 0.4	82.2 ± 0.7	74.6 ± 0.5	64.1 ± 0.6	62.8 ± 0.5						
M-LHRA (16)	57.1 ± 0.6	59.4 ± 0.4	21.5 ± 0.7	43.8 ± 0.4	77.5 ± 0.7	73.4 ± 0.5	60.4 ± 0.7	61.9 ± 0.4	84.1 ± 0.7	77.0 ± 0.6	67.7 ± 0.6	67.9 ± 0.5						
M-LHRA (128)	58.1 ± 0.6	61.4 ± 0.4	22.1 ± 0.7	45.4 ± 0.4	78.5 ± 0.7	75.9 ± 0.5	61.4 ± 0.7	63.7 ± 0.4	84.9 ± 0.7	79.1 ± 0.5	68.7 ± 0.6	69.6 ± 0.5						
M-RA (2)	51.9 ± 0.6	53.6 ± 0.4	19.6 ± 0.6	39.4 ± 0.4	72.8 ± 0.7	66.9 ± 0.6	55.1 ± 0.6	56.0 ± 0.4	79.6 ± 0.7	70.5 ± 0.5	62.5 ± 0.6	60.6 ± 0.5						
M-RA (16)	56.2 ± 0.6	58.8 ± 0.4	21.2 ± 0.7	43.6 ± 0.4	77.5 ± 0.7	73.2 ± 0.5	59.8 ± 0.7	61.5 ± 0.4	83.6 ± 0.7	77.0 ± 0.5	67.0 ± 0.6	67.2 ± 0.5						
M-RA (128)	58.0 ± 0.6	61.5 ± 0.4	22.0 ± 0.7	45.1 ± 0.4	78.4 ± 0.7	75.8 ± 0.5	61.4 ± 0.7	63.6 ± 0.4	84.7 ± 0.7	79.0 ± 0.5	68.6 ± 0.6	69.4 ± 0.5						
V-LHRA (2)	57.5 ± 0.6	60.1 ± 0.4	22.4 ± 0.7	45.8 ± 0.4	76.2 ± 0.7	73.1 ± 0.6	60.1 ± 0.6	60.9 ± 0.5	83.3 ± 0.7	77.1 ± 0.5	66.6 ± 0.6	66.9 ± 0.5						
V-LHRA (16)	58.4 ± 0.6	61.4 ± 0.4	22.6 ± 0.7	46.6 ± 0.4	78.3 ± 0.7	75.4 ± 0.5	61.4 ± 0.7	63.4 ± 0.4	84.5 ± 0.7	78.9 ± 0.5	68.0 ± 0.6	69.1 ± 0.5						
V-LHRA (128)	58.5 ± 0.6	61.8 ± 0.4	22.7 ± 0.7	46.8 ± 0.4	78.7 ± 0.7	76.2 ± 0.6	61.6 ± 0.7	64.2 ± 0.4	84.8 ± 0.7	78.9 ± 0.5	68.3 ± 0.6	70.3 ± 0.5						
V-RA (2)	55.9 ± 0.6	57.1 ± 0.4	21.6 ± 0.7	44.1 ± 0.4	73.0 ± 0.7	67.8 ± 0.5	58.2 ± 0.6	58.5 ± 0.5	80.6 ± 0.7	71.4 ± 0.6	63.8 ± 0.6	64.0 ± 0.5						
V-RA (16)	58.2 ± 0.6	60.8 ± 0.4	22.5 ± 0.7	46.2 ± 0.4	77.8 ± 0.7	73.6 ± 0.5	61.4 ± 0.7	62.6 ± 0.4	84.4 ± 0.7	77.0 ± 0.5	67.5 ± 0.6	68.8 ± 0.5						
V-RA (128)	58.4 ± 0.6	61.9 ± 0.4	22.6 ± 0.7	46.8 ± 0.4	78.6 ± 0.7	76.1 ± 0.6	61.4 ± 0.7	64.0 ± 0.4	84.6 ± 0.7	78.7 ± 0.5	68.2 ± 0.6	70.0 ± 0.5						

Table 3 continued

Distribution	2C		2C		2C		3C		3C		3C		U		U		U	
	0.5	Small	0.75	Small	0.5	Large	0.75	Small	0.75	Small	0.5	Large	0.5	Small	0.75	Small	0.5	Large
DOD																		
Area size																		
$\delta_{\max} = 3$																		
M	84.1 ± 0.5	77.8 ± 0.3	71.2 ± 0.4	61.5 ± 0.3	80.1 ± 0.4	72.7 ± 0.3	64.4 ± 0.4	53.5 ± 0.4	61.6 ± 0.4	50.0 ± 0.4	24.8 ± 0.7	18.4 ± 0.5						
VFA	82.0 ± 0.5	77.1 ± 0.3	71.2 ± 0.4	63.3 ± 0.4	76.1 ± 0.5	73.4 ± 0.4	64.5 ± 0.4	55.7 ± 0.3	63.7 ± 0.4	54.2 ± 0.3	28.0 ± 0.7	22.0 ± 0.5						
M-LHRA (2)	76.9 ± 0.5	74.2 ± 0.3	69.2 ± 0.4	61.3 ± 0.3	74.3 ± 0.5	69.7 ± 0.4	63.3 ± 0.4	55.2 ± 0.3	61.7 ± 0.4	52.7 ± 0.3	29.9 ± 0.8	23.3 ± 0.6						
M-LHRA (16)	83.0 ± 0.5	76.4 ± 0.3	70.9 ± 0.4	63.3 ± 0.3	78.5 ± 0.4	71.6 ± 0.3	65.2 ± 0.4	58.2 ± 0.3	64.0 ± 0.4	56.2 ± 0.3	32.5 ± 0.8	28.2 ± 0.6						
M-LHRA (128)	85.0 ± 0.5	79.7 ± 0.3	74.3 ± 0.4	67.5 ± 0.4	82.0 ± 0.4	75.6 ± 0.3	68.5 ± 0.4	61.3 ± 0.3	66.7 ± 0.4	58.9 ± 0.3	34.0 ± 0.9	30.7 ± 0.6						
M-RA (2)	75.3 ± 0.5	64.1 ± 0.3	62.1 ± 0.4	51.9 ± 0.3	69.6 ± 0.4	58.4 ± 0.3	55.6 ± 0.4	46.4 ± 0.3	52.6 ± 0.3	42.8 ± 0.3	25.7 ± 0.6	18.2 ± 0.4						
M-RA (16)	82.9 ± 0.5	72.9 ± 0.3	69.9 ± 0.4	60.1 ± 0.3	78.3 ± 0.4	67.5 ± 0.3	62.7 ± 0.4	54.3 ± 0.3	60.5 ± 0.4	51.0 ± 0.3	30.6 ± 0.8	24.6 ± 0.6						
M-RA (128)	84.9 ± 0.5	79.6 ± 0.3	74.3 ± 0.4	67.4 ± 0.3	81.9 ± 0.4	75.0 ± 0.3	68.6 ± 0.4	61.1 ± 0.3	66.5 ± 0.4	58.2 ± 0.3	33.7 ± 0.9	30.6 ± 0.6						
V-LHRA (2)	78.0 ± 0.5	69.2 ± 0.4	67.6 ± 0.5	62.0 ± 0.3	75.5 ± 0.4	66.4 ± 0.4	63.9 ± 0.4	56.1 ± 0.3	62.4 ± 0.4	54.2 ± 0.3	29.2 ± 0.8	20.6 ± 0.5						
V-LHRA (16)	84.6 ± 0.5	79.2 ± 0.3	73.6 ± 0.4	67.2 ± 0.3	81.3 ± 0.4	75.5 ± 0.4	68.0 ± 0.4	60.9 ± 0.3	66.5 ± 0.4	59.3 ± 0.2	32.7 ± 0.8	27.6 ± 0.6						
V-LHRA (128)	85.0 ± 0.5	80.4 ± 0.3	74.7 ± 0.4	67.7 ± 0.3	82.1 ± 0.4	76.7 ± 0.3	69.1 ± 0.4	62.3 ± 0.3	67.5 ± 0.4	60.4 ± 0.3	34.6 ± 0.9	31.9 ± 0.6						
V-RA (2)	79.0 ± 0.5	67.4 ± 0.3	65.8 ± 0.4	56.0 ± 0.4	76.3 ± 0.4	61.7 ± 0.3	59.3 ± 0.4	49.8 ± 0.3	57.5 ± 0.4	47.2 ± 0.3	27.6 ± 0.7	18.0 ± 0.4						
V-RA (16)	84.6 ± 0.5	78.1 ± 0.3	73.4 ± 0.4	64.9 ± 0.3	81.2 ± 0.4	72.9 ± 0.3	67.0 ± 0.4	58.1 ± 0.3	65.5 ± 0.4	56.7 ± 0.3	32.4 ± 0.8	25.5 ± 0.6						
V-RA (128)	85.0 ± 0.5	80.3 ± 0.3	74.6 ± 0.4	68.0 ± 0.3	82.1 ± 0.4	76.4 ± 0.3	69.1 ± 0.4	61.8 ± 0.3	67.3 ± 0.4	59.9 ± 0.3	34.3 ± 0.9	31.1 ± 0.6						

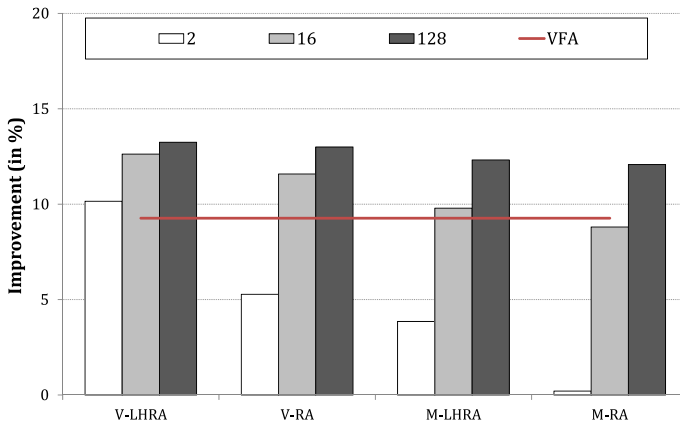


Fig. 2 Average improvement of the policies for $\delta_{\max} = 1$

For policies with online simulation runs, the number of simulation runs is depicted in the brackets. The proposed policy V-LHRA with 128 simulation runs achieves the best results in 19 of the 24 instance settings. Policy V-RA with 128 simulation runs achieves relatively good results as well. This indicates that 128 simulation runs are able to reflect a large portion of potential MDP-developments over the entire horizon. For 128 simulation runs, limiting the horizon is therefore not always necessary. However, V-RA with 128 simulation runs requires significantly more runtime compared to V-LHRA as shown in detail in Sect. A.4 of the “Appendix”. Because customers or drivers are waiting for response, these runtimes may not be applicable in practice. When the number of simulation runs is reduced to 16 (or even 2), the differences between V-LHRA and V-RA are significant. This indicates that limiting the online horizon and using the VFA to capture the remainder of the horizon becomes necessary when the number of simulation runs is limited. We analyze this phenomenon in the following in more detail. We present the results for the single-period case $\delta_{\max} = 1$ and then focus on the multi-period instances with $\delta_{\max} = 3$.

In the following, we compare the results for the single-period version of the MDRPSR in detail. To this end, we calculate the improvement of the policies compared to the myopic policy π^m :

$$\frac{Q(\pi, i) - Q(\pi^m, i)}{Q(\pi^m, i)}. \quad (6)$$

We list the improvement per instance setting in Table 4 of the Appendix. The average improvement over all instance settings is depicted in Fig. 2. The x-axis shows the RA-policies for the different number of online simulation runs $m = 2, 16, 128$. The y-axis shows the improvement compared to the myopic policy. The line indicates the improvement of the VFA.

V-LHRA outperforms all other policies for every number of online simulations achieving an improvement of up to 13.3% compared to the myopic policy. This means that on average 13.3% more customers can be served the same day they request.

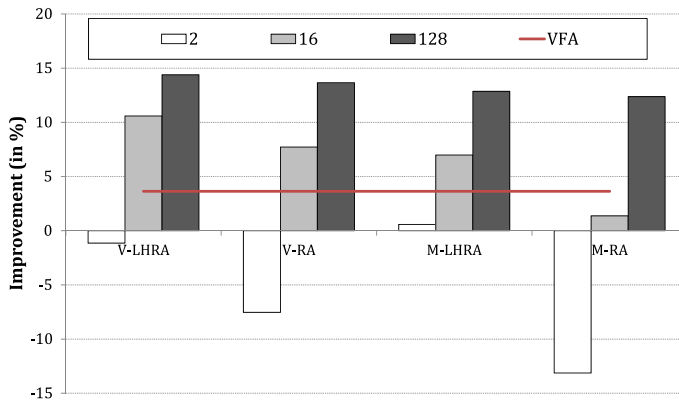


Fig. 3 Average improvement of the policies for $\delta_{\max} = 3$

V-LHRA therefore outperforms the current best policy for the single-period case, V-RA by Ulmer et al. (2018a). The improvement of a limited horizon is in particular high for a smaller number of samples. Even for a small number of $m = 2$ simulations per decision point, V-LHRA is able to achieve better results than the underlying offline VFA. With 16 simulation runs, V-LHRA achieves better results than M-RA with 128. The combination of limited horizon RAs and offline VFA therefore allows either a significant reduction of required online simulations and/or higher quality solutions for the same number of simulation runs. This is particularly important in dynamic settings with limited calculation time. Notably, the difference between V-LHRA and V-RA decreases for increasing simulation runs. This means that for this problem with $\delta_{\max} = 1$, reasonable results are possible by simulating until the end of the horizon with $m \geq 16$ simulation runs. This changes for the longer horizons of the multi-period settings, depicted in Fig. 3. Policy V-LHRA again achieves the highest solution quality. The improvement of V-LHRA reaches even up to 14.4%. For simulation runs $m \geq 16$, policy V-LHRA significantly outperforms the current state-of-the-art policy, the VFA by Ulmer et al. (2018c). However, the complexity of the sub-MDP increases and a low number of two simulation runs does not provide improvement compared to the myopic policy. In comparison to the single-period case, V-RA achieves less improvement for $m = 16$ simulation runs. As expected, simulating until the end of the horizon impedes the approximation when the overall problem horizon increases. This finding confirms that limiting the horizon is particularly important when the online calculation time is limited.

5.3 Impact of limited horizon

In this final section, we analyze the limited horizon and how it impacts the approximation and solution quality as well as the runtime. We focus on the multi-period case. We first analyze how the best horizon develops with respect to the number of

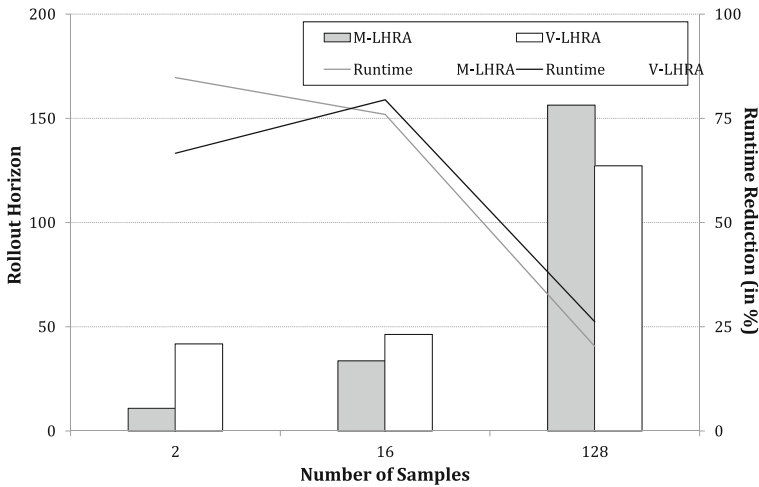


Fig. 4 Best horizon and runtime reduction per number of samples, $\delta_{\max} = 3$

samples. The bars in Fig. 4 depicts the average number of horizon steps h with respect to the base policy and the number of samples. The horizon is depicted on the left y-axis. For both policies V-LHRA and M-LHRA, a tendency can be observed. The higher the number of samples, the larger is the simulation horizon of the best policy. The tendency for V-LHRA is milder compared to M-LHRA. This can be explained by the offline VFA reinforcing the online approximation. Generally speaking, longer simulation horizons are suitable in cases the computational resources allow a larger number of simulation runs. This confirms the observation by Ghiani et al. (2009) and Voccia et al. (2017). If the number of online simulations is small, shorter horizons are advantageous. As we show in the following, the results for different horizons of the V-LHRA do not vary that significantly compared to the results of the M-LHRA.

Since all simulations are conducted in real-time, we now analyze the runtime reduction gained by the limited horizon RAs. Since the runtimes differ per decision point, we focus on the runtime at the decision point per realization with the maximal required runtime. Hence, we compare the average maximal runtime per decision point per realization. To this end, we calculate the average maximal runtime over all instance settings per policy. The individual values are depicted in the “Appendix”. The reduction is the decrease of runtime compared to the runtime of the full-horizon RAs, M-RA and V-RA respectively. The reduction is indicated by the lines in Fig. 4. The percentage of reduction is depicted on the right y-axis. We observe a strong correlation to the horizon h . If the horizon is long, the reduction is less distinct. However, even for 128 simulation runs and an average V-LHRA horizon of 156 decision points, the reduction is significant with 26.2%. For 16 simulation runs and V-LHRA, the average maximal runtime per decision point is reduced from 538.4 s to 110.8 s. The reduction from nearly 9 min to less than 2 min may therefore enable real-time implementation by even higher solution quality compared to V-RA.

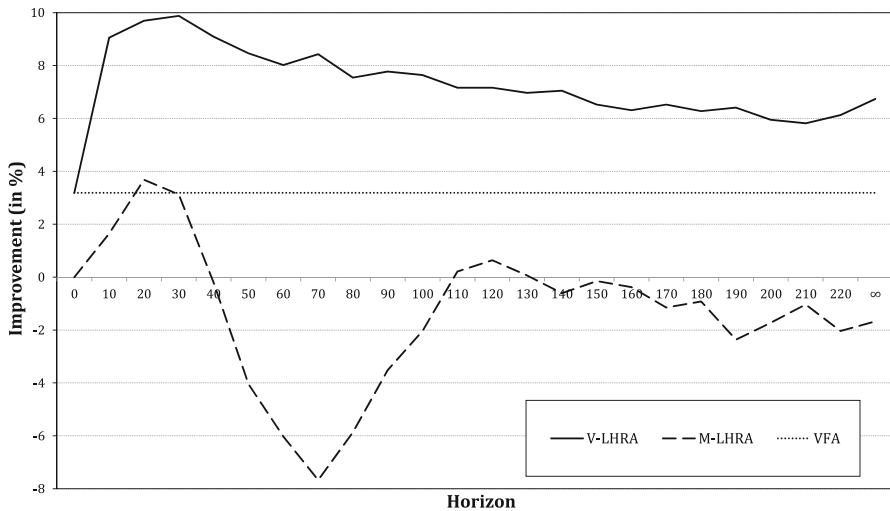


Fig. 5 Improvement for 2C, large service area, DOD of 0.75, $\delta_{\max} = 3$, 16 simulations

Our experiments also indicate that the number of simulation runs and the corresponding horizon should depend on the runtime available. In the vast majority of instance settings, V-LHRA with 128 simulation runs and a very short horizon of only 10 or 20 decision points outperforms the best tunings of V-LHRA with 2 and 16 simulations runs. This indicates that given limited runtime the priority should be on the number of simulation runs first before increasing the horizon lengths.

We finally analyze the impact of the horizon for one instance setting. We select instance setting 2C, large service area, DOD of 0.75 since it requires the highest runtimes. However, the general tendency is similar for most of the instance settings. We apply the RAs with 16 simulation runs. The development of the improvement compared to the myopic policy for V-LHRA and M-LHRA over differing horizons is depicted in Fig. 5. The horizon is depicted on the x-axis. A horizon of 0 reflects the base policy. A horizon of ∞ represents V-RA and M-RA. Analyzing V-LHRA, we observe an increase until $h = 30$ followed by a slow, but constant decrease. For this instance setting, the detail accuracy of the VFA-based RA lasts until 30 decision points. After these, the reliable approximation via offline VFA is superior.

The importance of the offline VFA to approximate the end of the horizon becomes evident when we compare the results with M-LHRA. First, the results are generally inferior to V-LHRA. The RA with myopic base policy additionally shows a different behavior over the horizon length. The values increase until $h = 20$, then drop drastically and start to recover around $h = 70$. With 70 decision points, the impact of the decision on the next period starts to reflect in the evaluation. Before this horizon length, the RA mainly captures the developments in the current period but ignores the impact of postponements. The V-LHRA incorporates the development of future periods by drawing on the offline VFA and therefore provides significant improvement relatively independent of the horizon length.

6 Conclusion and outlook

In this paper, we have presented a novel combination of approximate dynamic programming methods utilizing the advantages of offline and online ADP. We tested our method for two dynamic vehicle routing problems from the literature. Both problems are represented by a general model, the MDPRSR. Our generic approach combines benefits of online simulation and offline value function approximation and is especially suitable for dynamic vehicle routing problems with a large number of decision points. For both the single- and multi-period version of the MDPRSR, the presented approach significantly outperforms state-of-the-art benchmark policies while additionally reducing the required online calculation time drastically.

Future research may address both the MDPRSR and the presented methodology. The MDPRSR may be extended to multiple vehicles. Here, the challenge would be in particular to derive a suitable offline VFA. Another interesting avenue could be to increase the number of periods, or even extend the MDPRSR to an infinite horizon problem. Further, as our results indicate, runtime should be invested first in the number of simulation runs and then in the horizon lengths. Future research may analyze the suitable allocation in number of simulation runs and lengths of simulations. Finally, the proposed method may be generalized by learning suitable horizons dependent on the states. To this end, measures such as the indifference zone selection by Kim and Nelson (2006) may be extended to determine suitable horizons and simultaneously reduce the required number of simulations.

Appendix

In the Appendix, we present describe the routing heuristic and the detailed algorithm for the V-LHRA. We further describe how the customer distributions are generated and finally provide the results of the tuning and runtimes in detail.

A.1 Routing heuristic

The routing is conducted with a cheapest insertion procedure (CI). The tour is initialized and updated regarding the following procedure: In $\delta = 1$ and $t = 0$, CI starts with a pre-decision tour $\theta_0 = (D, D)$ only consisting of the depot. In every decision point k , a pre-decision tour θ_k^δ and a set of candidate subsets $\hat{C}_k^a \subseteq C_k^r, i = 1, \dots, 2^{|C_k^r|}$, i.e., the power set of new requests is given. For every candidate subset, CI subsequently selects the cheapest request of the subset regarding the insertion time and adds it at the cheapest insertion position in the tour. The same procedure is applied for $\theta_k^{\delta+1}$ with the postponed requests $C_k^r \setminus \hat{C}_k^a$. The routing is induced by the accepted subset $\hat{C}_k^a$ and the according post-decision tour $\theta_k^{\delta,x}$. If the selected routing decision of x is not waiting, the first customer in the tour $C_{\text{next}} = \theta_1$ is the next customer to visit. After the travel to the next customer, the visited customer is removed from $\theta_k^{\delta,x}$ resulting in pre-decision tour θ_{k+1}^δ . Waiting is applied if no customers are left to serve, i.e., $\theta_k^\delta = (\mathcal{P}_k, \mathcal{D})$, but

free time budget is left. In this case, the pre-decision tour $\theta\delta_{k+1}$ is identical to $\theta_k^{\delta,x}$. If no time is left, the vehicle returns to the depot. A candidate subset and the according decision are considered feasible if the resulting candidate post-decision tour does not exceed the time limit. Because the decision space is reduced, CI may reject candidate subsets, which are feasible regarding a different routing approach, e.g., an optimal traveling salesman solution.

A.2 V-LHRA algorithm

Algorithm 1 describes the procedure of decision making for a VFA-based limited horizon post-decision RA. Let a state S_k , decisions $x_1, \dots, x_n$, the reward-function R , the PDSs $\mathcal{P}_k = (S_k^{x_1}, \dots, S_k^{x_n})$, a set of sampled realizations $\{\omega_1, \dots, \omega_m\}$, the

Algorithm 1: VFA-Based Limited Horizon Rollout Algorithm

Data: State S_k , Decisions $\{x_1, \dots, x_n\}$, Reward-Function $R(S, x) \rightarrow \mathbb{N}_0$, PDSs $S_k^x = (S_k^{x_1}, \dots, S_k^{x_n})$, Realizations $\{\omega_1, \dots, \omega_m\}$, Base Policy π_{VFA} , VFA $\tilde{V}$, Horizon h
Result: Decision x^*

```

1 forall the  $S^x \in S_k^x$  // For All Potential Decisions
2 do
3    $i \leftarrow 0$ 
4    $\hat{V}(S^x) \leftarrow 0$ 
5   // Online Simulation
6    $j \leftarrow 0$ 
7   while  $(i \leq m)$  // For Each Realization
8   do
9      $S \leftarrow S_k$ 
10     $S^x \leftarrow S_k^x$ 
11    while  $(j \leq h)$  // Simulate the Limited Horizon
12    do
13       $S \leftarrow (S^x, \omega_i)$ 
14       $x \leftarrow X^{\pi_{\text{VFA}}}(S)$  // Apply VFA Base Policy
15       $\hat{V}(S_k^x) \leftarrow \hat{V}(S_k^x) + \frac{1}{m} R(S, x)$  // Collect Simulated Rewards
16       $S^x \leftarrow (S, x)$ 
17       $j \leftarrow j + 1$ 
18    end
19    // Add Offline Value
20     $\hat{V}(S_k^x) \leftarrow \hat{V}(S_k^x) + \frac{1}{m} \tilde{V}(S^x)$ 
21     $i \leftarrow i + 1$ 
22  end
23  // Select Decision with Bellman Equation
24   $R^* \leftarrow 0$ 
25  forall the  $S_k^x \in \mathcal{P}_k$  do
26    if  $(R(S_k, x) + \hat{V}(S_k^x) \geq R^*)$  then
27       $x^* \leftarrow x$ 
28       $R^* \leftarrow R(S_k, x) + \hat{V}(S_k^x)$ 
29    end
30  end
31 end
32 return  $x^*$ 

```

VFA base policy π_{VFA} , and the VFA-values $\tilde{V}$ be given. Then, for every PDS S_k^x , the RA simulates the next h decision points of every realization ω_i . In the decision points, the decision $X^{\pi_{\text{VFA}}}(S_{k+j})$ induced by the VFA base policy π_{VFA} is applied and the rewards $R(S_{k+j}, X^{\pi_{\text{VFA}}}(S_{k+j}))$ are accumulated. After the h decision points, the value $\tilde{V}(S_{k+h}^x)$ is added to the accumulated rewards. The overall reward-to-go of a PDS $\hat{V}(S_k^x)$ is the average of the single rewards per realization. The RA selects the decision x^* leading to the maximum sum of immediate reward $R(S_k, x^*)$ and expected future rewards $\hat{V}(S_k^{x^*})$.

A.3 Customer distributions

In this section, we describe the spatial customer distributions for small service area size. For the large service area, the quantities are multiplied by 1.5. Given U , a spatial realization (x, y) is defined as $(x, y) \sim U[0, 10] \times U[0, 10]$. For $2C$, the customers are equally distributed to each cluster. The cluster centers are located at $\mu_1 = (2.5, 2.5)$, $\mu_2 = (7.5, 7.5)$. The standard deviation within the clusters is $\sigma = 0.5$. For $3C$, the cluster centers are located at $\mu_1 = (2.5, 2.5)$, $\mu_2 = (2.5, 7.5)$, $\mu_3 = (7.5, 5)$. 50% of the requests are assigned to cluster two, 25% to each other cluster. The standard deviations are set to $\sigma = 0.5$.

A.4 Results

In this section, we present the best simulation horizons and the maximal runtimes per decision point per instance setting.

See Tables 4, 5, 6, 7 and 8.

Table 4 Results: improvement compared to Myopic (in %)

Distribution	Area size	DOD	VFA	M-LHRA			M-RA			V-LHRA			V-RA		
				2	16	128	2	16	128	2	16	128	2	16	128
Simulation runs															
$\delta_{\max} = 1$															
2C	Small	0.5	-0.8	-1.2	1.2	1.8	-4.2	0.6	1.9	0.2	1.7	1.9	-3.0	1.6	1.8
2C	Small	0.75	-0.7	-2.6	0.5	3.2	-8.0	0.5	3.1	0.6	2.9	3.0	-6.8	0.5	2.7
2C	Large	0.5	3.3	1.9	7.7	9.3	-0.4	6.7	9.2	5.9	8.0	8.6	1.6	7.4	8.6
2C	Large	0.75	5.1	-1.2	6.9	9.4	-4.6	5.7	9.2	5.3	8.8	10.3	0.8	8.3	10.2
3C	Small	0.5	-1.4	-0.4	2.6	4.1	-3.5	2.8	4.0	1.1	3.8	4.3	-3.2	3.2	4.3
3C	Small	0.75	-0.5	-1.7	1.8	5.4	-7.1	1.6	5.2	1.6	4.7	5.7	-5.9	2.2	5.6
3C	Large	0.5	7.7	4.3	9.4	11.3	-0.2	8.3	11.2	8.9	11.2	11.7	5.4	11.2	11.3
3C	Large	0.75	7.7	4.2	12.2	14.8	-0.2	9.6	13.4	10.4	14.8	16.2	4.2	11.6	14.0
U	Small	0.5	14.3	6.9	12.9	14.9	2.5	11.2	14.6	13.5	15.2	15.7	10.5	15.1	15.4
U	Small	0.75	11.1	3.0	10.2	13.8	-0.5	9.0	14.1	11.5	13.9	14.7	5.9	12.8	14.8
U	Large	0.5	39.2	22.9	32.6	36.2	21.3	30.8	36.2	38.2	39.3	39.9	33.6	39.3	39.6
U	Large	0.75	26.2	10.0	19.5	23.6	7.3	18.9	23.0	24.8	27.1	27.0	20.2	25.9	27.6

Table 4 continued

Distribution	Area size	DOD	VFA	M-LHRA			M-RA			V-LHRA			V-RA		
				2	16	128	2	16	128	2	16	128	2	16	128
Simulation runs															
$\delta_{\max} = 3$															
2C	Small	0.5	-2.5	-8.6	-1.3	1.0	-10.3	-1.5	1.1	-7.2	0.6	1.1	-6.2	0.7	1.1
2C	Small	0.75	-1.0	-4.6	-1.9	2.3	-17.8	-6.5	2.0	-11.3	1.8	3.3	-13.3	0.5	3.1
2C	Large	0.5	0.0	-2.7	-0.4	4.7	-12.7	-1.7	4.7	-5.1	3.7	5.1	-7.5	3.2	5.1
2C	Large	0.75	3.2	0.1	3.7	11.1	-15.0	-1.7	9.8	1.4	9.9	11.5	-7.9	6.7	11.3
3C	Small	0.5	-5.0	-7.2	-2.0	2.4	-12.9	-2.1	2.2	-5.7	1.5	2.3	-4.6	1.4	2.5
3C	small	0.75	1.0	-4.1	-1.4	4.0	-19.7	-7.2	3.3	-8.6	3.9	5.5	-15.2	0.2	5.3
3C	Large	0.5	0.1	-1.8	1.2	6.4	-13.7	-2.6	6.4	-0.9	5.5	7.3	-8.1	4.0	7.2
3C	Large	0.75	4.3	3.3	8.9	14.0	-13.1	1.3	14.4	4.8	13.8	16.6	-7.0	8.9	15.5
U	Small	0.5	3.4	0.1	3.9	8.4	-14.4	-1.9	8.2	1.4	8.0	9.6	-6.6	6.1	9.0
U	Small	0.75	8.3	-14.4	-11.5	-4.4	-30.7	-17.4	-5.5	-11.9	-3.7	-1.9	-23.3	-8.0	-2.7
U	Large	0.5	12.6	20.2	31.0	36.8	3.2	23.5	35.8	17.5	31.8	38.8	10.9	30.3	37.8
U	Large	0.75	19.4	26.7	53.6	67.8	-0.7	34.2	66.2	11.9	50.4	73.6	-1.8	38.7	68.7

Table 5 Results: best horizon, $\delta_{\max} = 1$

Distribution	Area size	DOD	M-LHRA			V-LHRA		
			2	16	128	2	16	128
Simulation runs								
2C	Small	0.5	5	50	45	5	15	30
2C	Small	0.75	5	70	70	5	10	75
2C	Large	0.5	35	40	50	15	55	60
2C	Large	0.75	10	35	75	5	35	65
3C	Small	0.5	5	80	75	10	20	70
3C	Small	0.75	5	70	55	5	15	40
3C	Large	0.5	20	35	75	10	20	75
3C	Large	0.75	20	45	65	10	25	75
U	Small	0.5	20	35	45	10	20	70
U	Small	0.75	15	30	45	5	25	80
U	Large	0.5	70	30	55	5	5	40
U	Large	0.75	15	20	65	5	10	30

Table 6 Results: best horizon, $\delta_{\max} = 3$

Distribution	Area size	DOD	M-LHRA			V-LHRA		
Simulation runs			2	16	128	2	16	128
2C	Small	0.5	10	210	210	220	220	220
2C	Small	0.75	10	10	220	40	20	130
2C	Large	0.5	10	10	210	30	30	150
2C	Large	0.75	10	20	130	30	30	140
3C	Small	0.5	10	210	210	210	210	100
3C	Small	0.75	10	10	140	20	20	150
3C	Large	0.5	10	10	130	20	40	110
3C	Large	0.75	10	20	110	20	20	80
U	Small	0.5	10	20	140	20	20	110
U	Small	0.75	10	20	130	20	20	110
U	Large	0.5	20	20	190	30	70	190
U	Large	0.75	10	20	110	20	30	130

Table 7 Results: maximal runtime per decision point (in s), $\delta_{\max} = 1$

Distribution	Area size	DOD	M-LHRA			M-RA			V-LHRA			V-RA		
			2	16	128	2	16	128	2	16	128	2	16	128
Simulation runs														
2C	Small	0.5	0.18	1.9	11.05	0.11	1.66	14.93	0.53	2.35	14.28	0.84	3.65	25.26
2C	Small	0.75	2.65	32.21	220.64	1.18	11.17	325.17	21.75	47.57	220.71	16.75	65.86	395.03
2C	Large	0.5	63.99	67.66	68.66	66.49	67.83	68.21	0.59	5.19	29.01	1.54	1.18	40.57
2C	Large	0.75	8.37	181.74	1513.1	7.55	146.38	1280.49	30.99	413.58	1366.88	32.71	221.83	1544.52
3C	Small	0.5	0.06	1.24	9.48	0.1	0.92	17.09	0.26	1.41	9.13	0.53	2.27	17.41
3C	Small	0.75	0.64	18.34	99.71	0.64	7.68	134.73	2.75	13.84	112.76	8.38	27.76	175.25
3C	Large	0.5	0.25	2.17	18.19	0.13	1.77	34.24	0.58	2.67	17.81	0.72	4.44	36.05
3C	Large	0.75	5.7	58.49	551.95	4.33	36.41	550.44	8.2	75.08	395.84	21.62	83.27	683.86
U	Small	0.5	0.17	1.56	12.39	0.12	1.34	26.61	0.55	1.95	14.35	0.57	3.7	28.52
U	Small	0.75	3.15	25.8	226.05	1.18	28.26	362.23	4.96	39.56	252.34	6.45	46.34	371.59
U	Large	0.5	0.25	2.79	13.76	0.16	0.61	29.21	0.28	2.01	13.91	0.67	5.3	38.57
U	Large	0.75	7.82	72.15	656.69	4.28	36.98	555.54	9.3	79.29	621.21	11.95	107.48	628.46

Table 8 Results: maximal runtime per decision point (in s), $\delta_{\max} = 3$

Distribution	Area size	DOD	M-LHRA			M-RA			V-LHRA			V-RA		
			2	16	128	2	16	128	2	16	128	2	16	128
Simulation runs														
2C	Small	0.5	1.2	60	529.8	9	66.6	473.4	12	95.4	727.8	10.8	85.2	740.4
2C	Small	0.75	15	81	3117	152.4	565.2	3516.6	79.2	118.2	3774.6	95.4	777.6	5388
2C	Large	0.5	2.4	68.4	787.8	17.4	84.6	1050.6	7.2	63.6	1182	20.4	133.8	1246.2
2C	Large	0.75	46.2	338.4	7602	187.8	1218	7607.4	100.2	469.8	7915.2	355.8	1990.2	10801.8
3C	Small	0.5	0.6	33.6	258	5.4	31.2	252.6	6	45	310.2	4.8	43.2	367.2
3C	Small	0.75	10.8	24	1738.8	40.8	253.2	1358.4	17.4	51.6	2047.8	42	247.8	2667.6
3C	Large	0.5	1.2	3.6	349.8	12	45	315.6	1.8	24.6	504	8.4	67.2	506.4
3C	Large	0.75	10.2	141.6	2899.8	83.4	555	3625.2	28.2	102	2855.4	110.4	555	4568.4
U	Small	0.5	0.6	5.4	277.8	4.2	33	241.8	1.2	6	279	5.4	46.2	355.2
U	Small	0.75	5.4	57	2052	54.6	415.2	2360.4	10.2	60.6	2292.6	69	1806.6	3953.4
U	Large	0.5	0.6	5.4	459.6	4.8	37.2	282	1.2	31.8	336	5.4	42.6	399.6
U	Large	0.75	7.8	100.2	486.6	97.8	516.6	4719	14.4	260.4	4937.4	108.6	678	5821.2

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